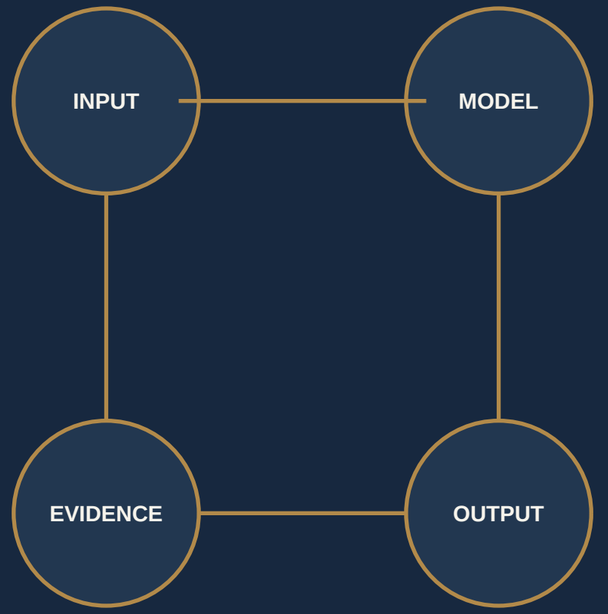
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Social Robotics

AI.312 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define social intelligence.
- Apply key methods including communication.
- Build or analyze a practical system involving emotion recognition.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

Social intelligence

02

METHODS

Communication

03

SYSTEMS

Emotion recognition

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · Social intelligence

Social intelligence

02 · Communication

Communication

03 · Emotion recognition

Emotion recognition

04 · Personality

Personality

05 · Trust

Trust

06 · Robot behavior and human-ce

Robot behavior and human-centered design

07 · Mandatory practical: Design a

Mandatory practical: Design and evaluate a social-robot interaction scenario

08 · ### Year 4 — Semester 1

Year 4 — Semester 1

Core ideas, made explicit

AI.312 • Distinguish definitions, assumptions and evidence

- Social intelligence
- Communication
- Emotion recognition
- Personality
- Trust

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

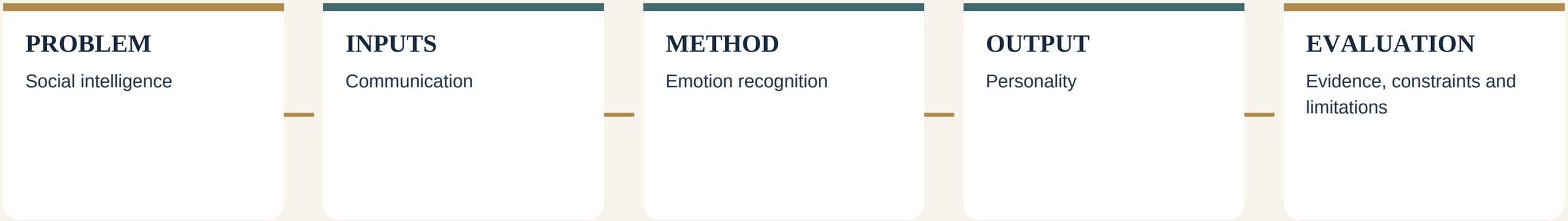
AI.312 • Distinguish definitions, assumptions and evidence

- Robot behavior and human-centered design
- Mandatory practical: Design and evaluate a social-robot interaction scenario
- ### Year 4 — Semester 1

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.



Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving social intelligence.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using social intelligence.
- Implement or simulate a baseline using communication.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.